Optical, electron microscopy and AFM

	Optical microscope	Scanning electron microscope	Transmission electron microscope	Atomic force microscope
Mechanism of contrast formation	Light absorption (diffraction, polarization, scattering etc) or fluorescence	Electron scattering	Electron absorption	Forces of probe-sample interaction
Range in XY	200nm – 1mm	1nm – 10 mm	1Å – 100μm	1Å – 200 μm
Range in Z	500nm – 10 mm	10 nm – 1 mm	limited by ultratome (10 nm – 1 μm)	0.5Å – 20 μm
What is studied	optical slice	surface	real thin slice	surface
Complicated sample preparation	optional	necessary	necessary	optional

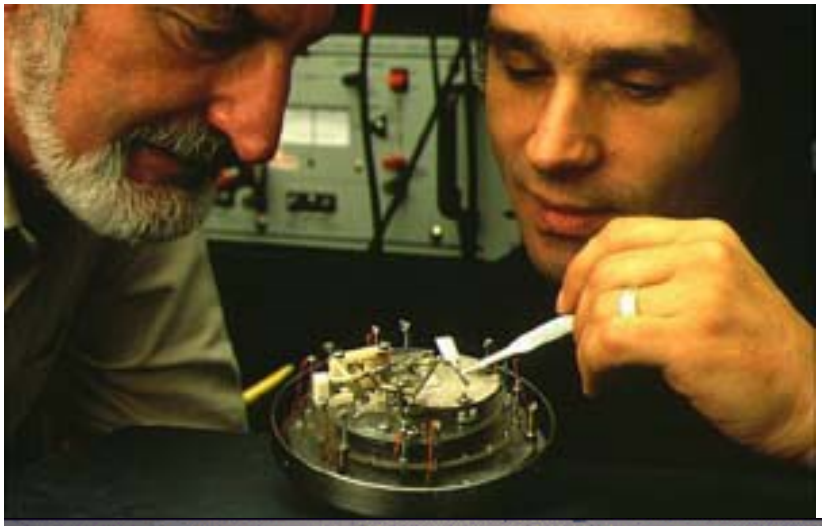
History points

First SPM was invented in 1981 by Heinrich Rohrer and Gerd K. Binnig, scientists at IBM's Zurich Research Laboratory in Switzerland (it was STM)

In 1986 Nobel Price was awarded for this invention

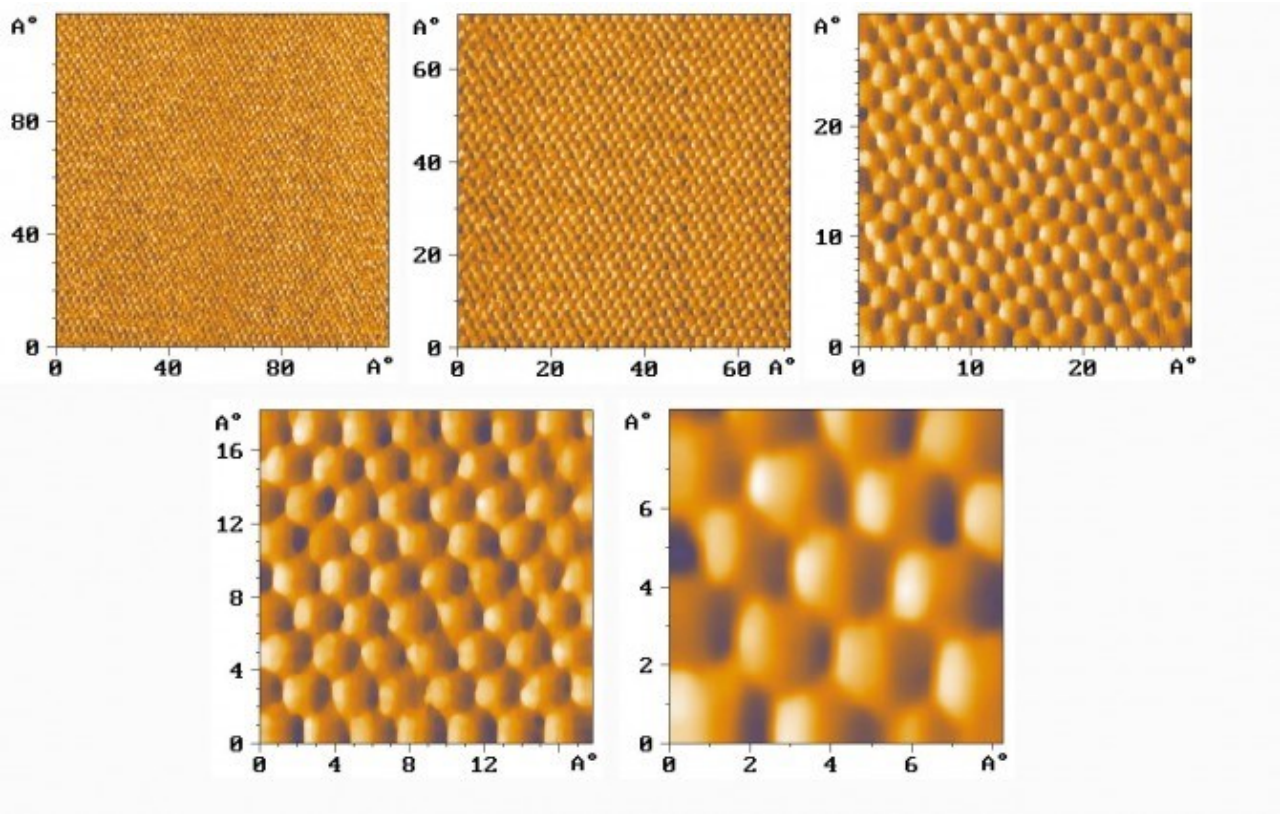
1966 – First STM

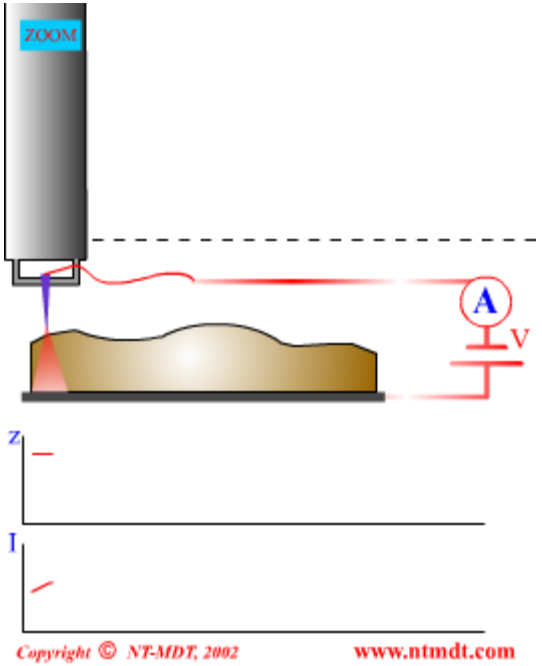
R. Young, J. Ward and F. Scire
"Topographer" NIST, USA





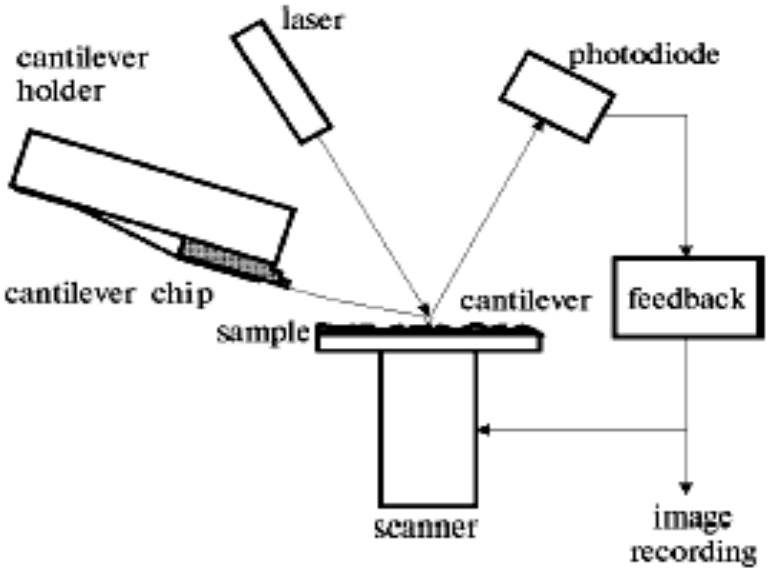
STM principle

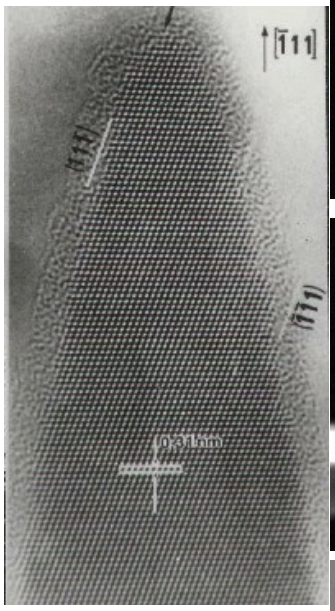


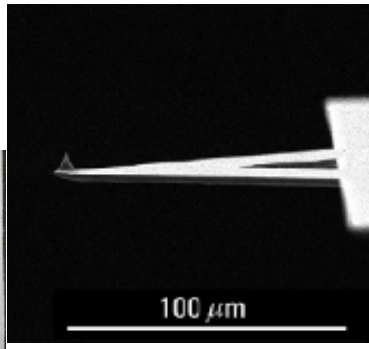


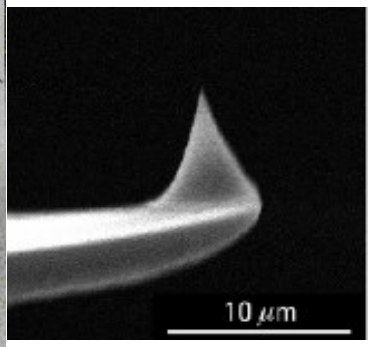
Tunneling current occurs at very sort distances when the voltage is applied between the probe and the sample. It is very sensitive to the tip-sample distance. This technique provides the best spatial resolution among other SPMs.

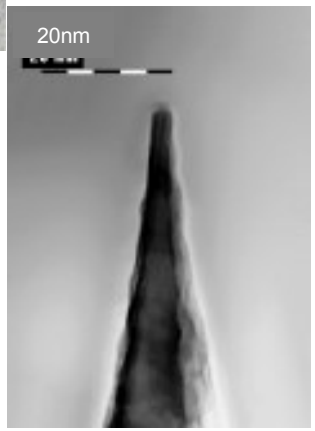
Contact AFM expands SPM application





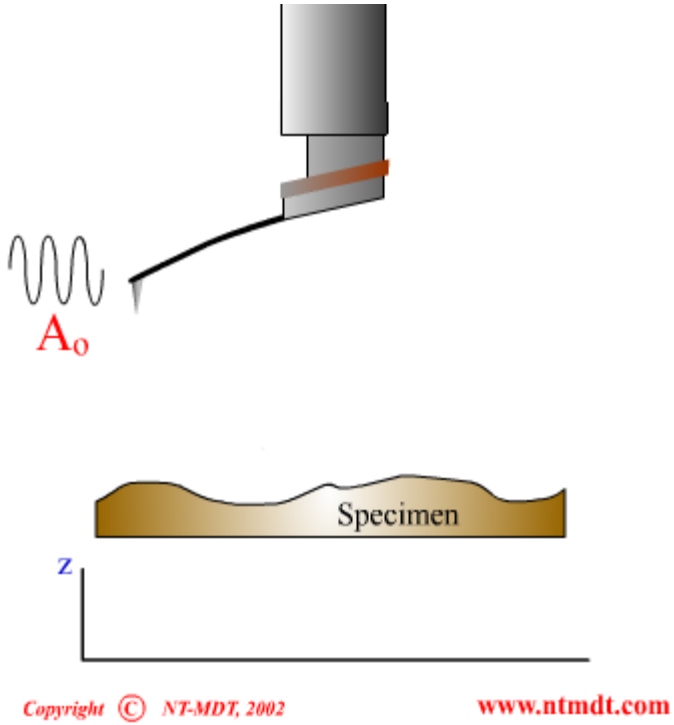






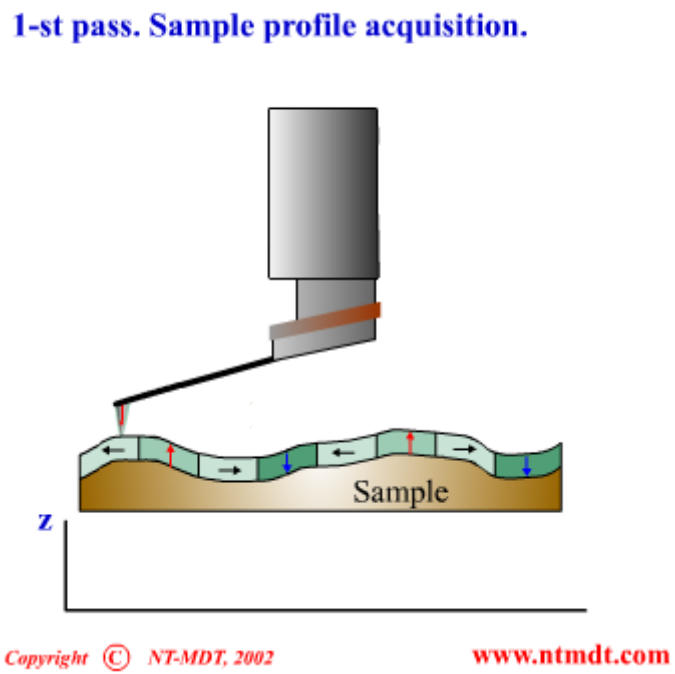
Sharp probe (as those on photos) placed very close to the sample surface moves along and its movement is registered by reflected laser beam. Different forces between the probe and the sample provide different modes of contrast formation.

Semi-Contact AFM



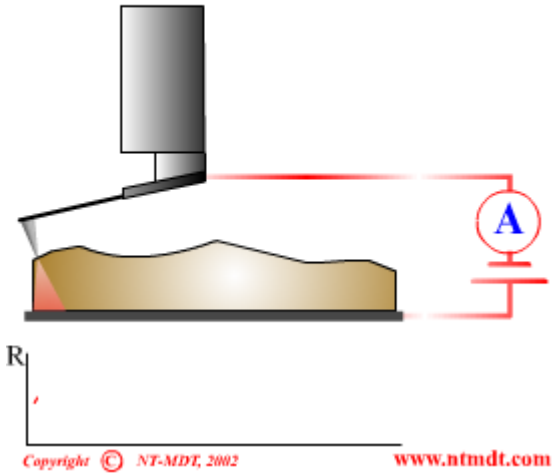
Semi-contact modes that are AFM techniques with the oscillating probe contacting (“touching”) the surface periodically in the extreme points of its trajectory. Oscillation frequency change crucially in close proximity to the surface. Point to point differences in oscillation frequency make an image contrast.

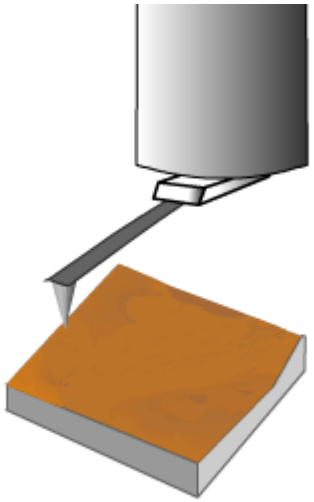
Many-Pass AFM techniques



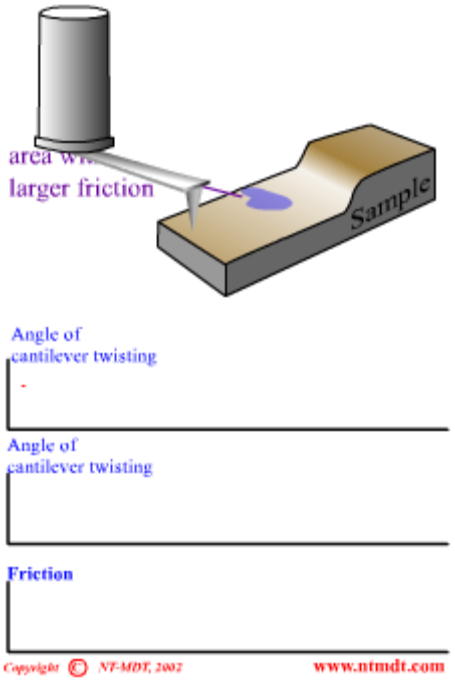
Methods for complex AFM characterization of the object. The first pass is performed in contact or semi-contact mode to determine the surface topology. By the next pass(es) additional information is obtained, for example, electrical, magnet or some other sample properties. Usually second pass scanning is performed in a non-contact mode when the probe follows the predetermined surface relief but moves a bit higher without touching the sample.

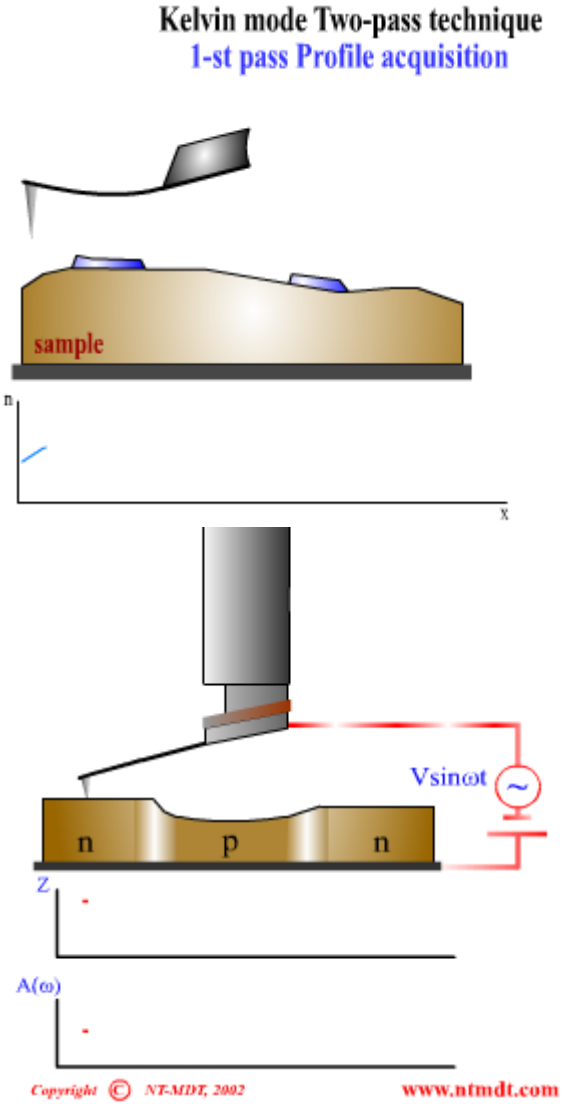
More than 40 measuring techniques





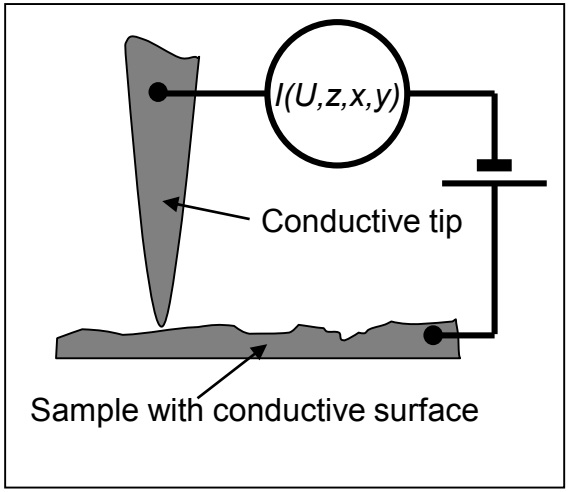
Lateral Force Microscopy
(LFM)

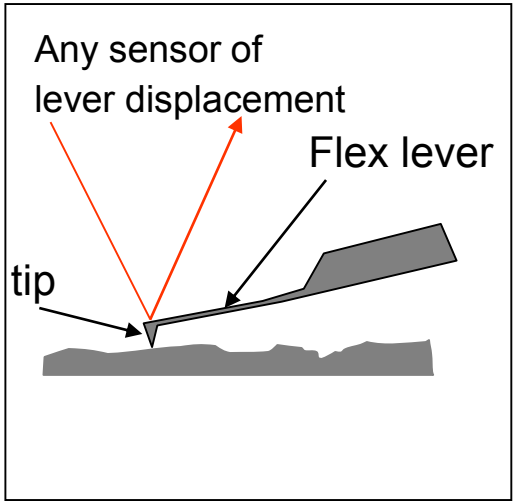


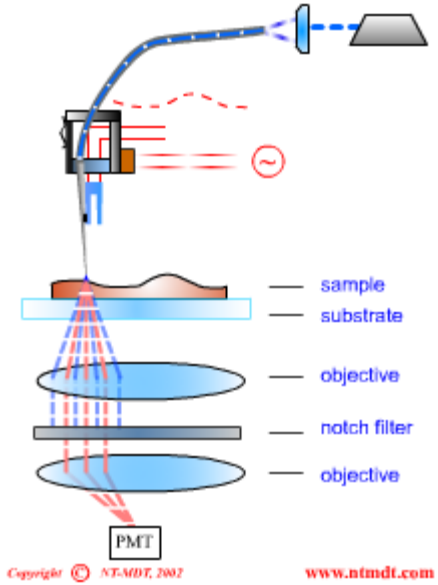


Scanning Probe Microscopy

Probe tip obtain the information from every point of an object surface. Different tips get different information.







Scanning Tunneling Microscopy

Conductive sample

Atomic Force Microscopy

Any solid surfaces

Scanning Near-Field Microscopy

Optical properties

Environmental control

SPM operation environments

- Air (controlled gas with humidity control)
- Liquid
- Electrochemical solutions
- Low vacuum (10^{-3} torr)
- High vacuum (10^{-7} torr)

Temperature control

- In air (-30C – 200C)
- In liquid (up to 60C)
- In high vaccum (20K-300K)

Attraction forces:

Far-distant: capillary and van-der-Waals (vdW),

Contact: adhesion forces due to the difference in probe/air and surface/air surface energies.

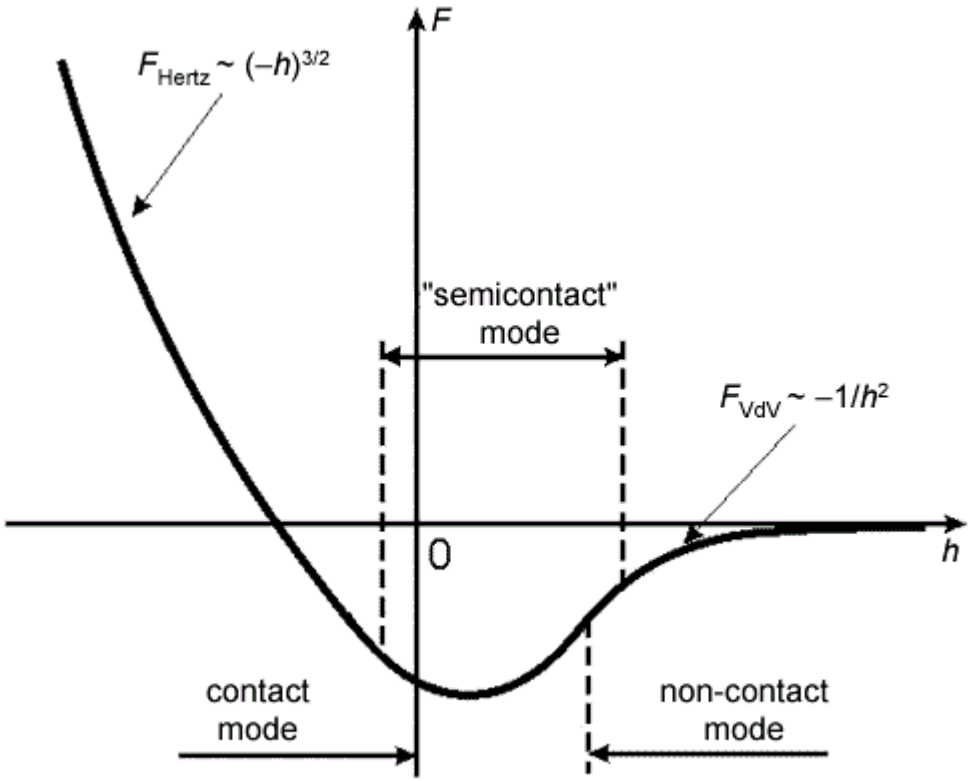
Repulsion forces:

Elastic (Hooke's) type,

vdW are repulsive, if medium's dielectric permittivity constant ε_m is intermediate between ε_t and ε_s correspondingly tip and sample.

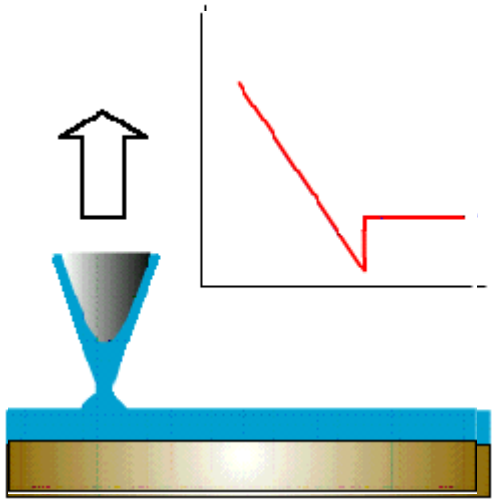
Electrostatic Coulomb forces can be both attractive and repulsive.

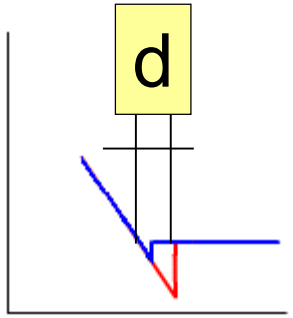
A general distance dependence of a force acting on a probe



Probe-sample interaction potential.

Force-distance curves





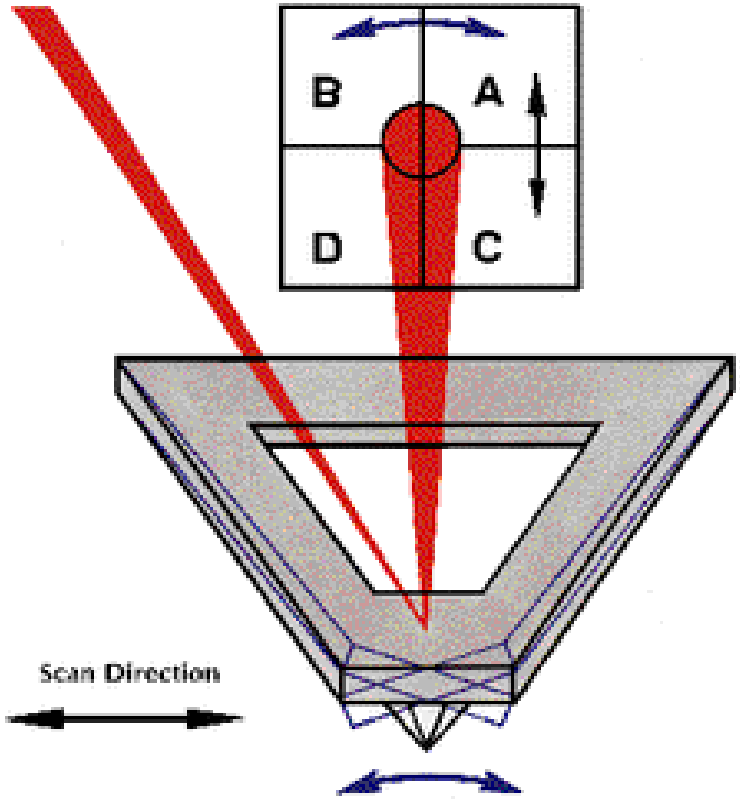
$$F_{adh} = k\,d$$

The adhesion results in a hysteresis in approach-retract deflection curves. The adhesion force is proportional to the extension of hysteresis area.

The condition for the hysterezis appearance in force-distance spectroscopy of a contact mode is:

$dF/dz > k_{cant}$

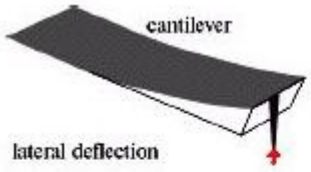
Lateral force imaging

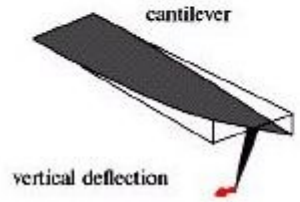


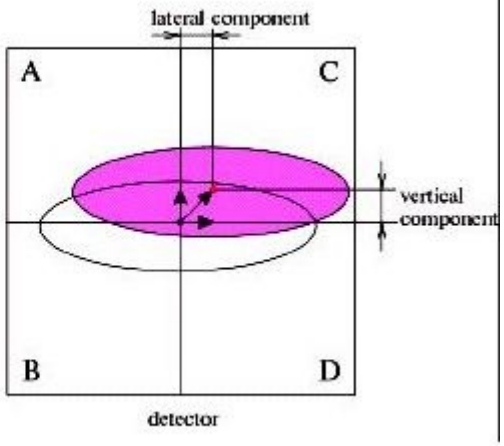
LF ~ (A+C)-(B+D)

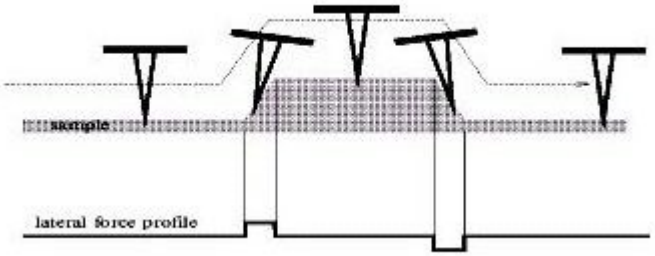
Lateral force imaging

At scanning in **constant force mode** *perpendicular to longitudinal axis* of the cantilever an additional torsional bending of the cantilever proportional to the side (lateral) force occurs along with the cantilever's deflection in the normal direction. It provides lateral force imaging in a contact mode.

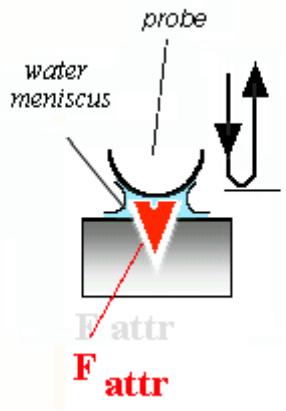






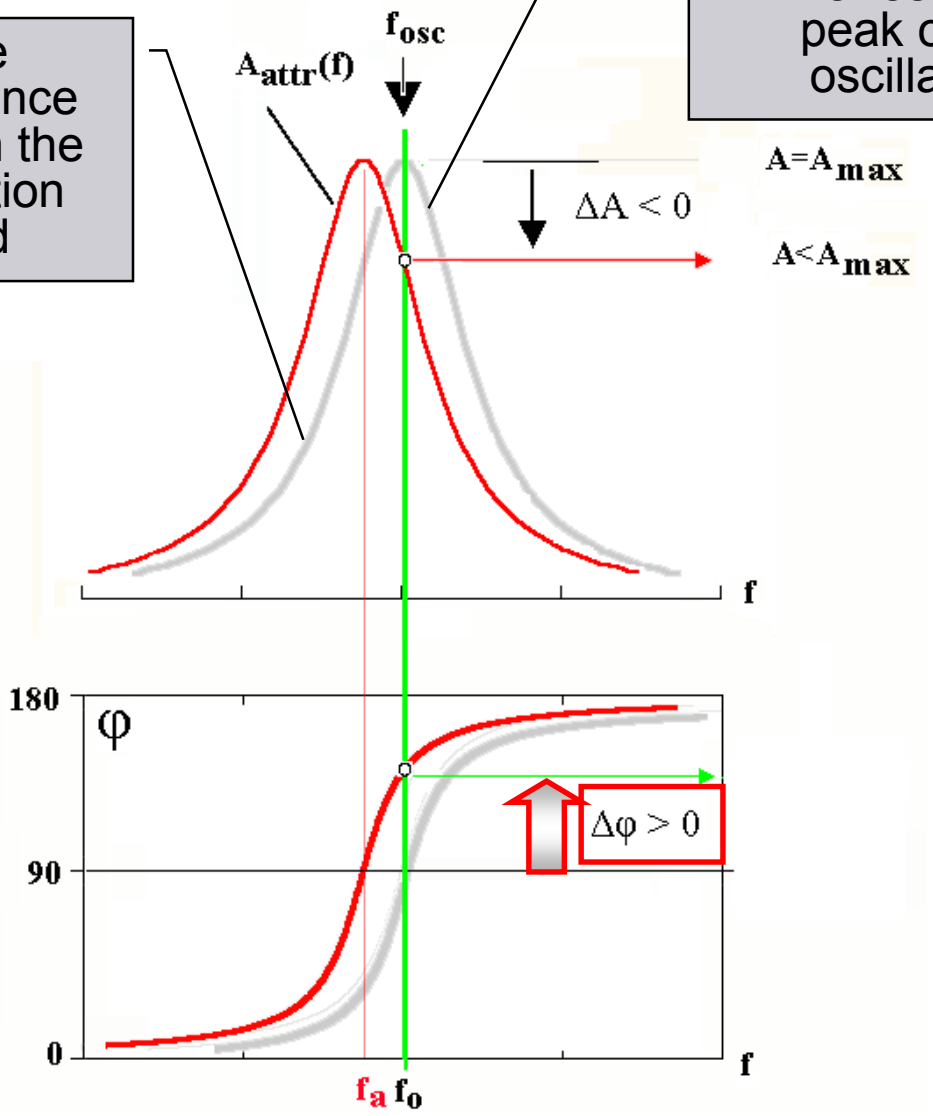


attraction



The resonance peak in the attraction field

The resonance peak of free oscillations



$k^* = k + \langle dF_{attr}/dz \rangle$,

$k^* < k$, $f_{attr} < f_0$

The resonance frequency shifts to the left,

The amplitude decreases,

The phase increases.

Advantages and Disadvantages of main AFM modes

Contact Mode

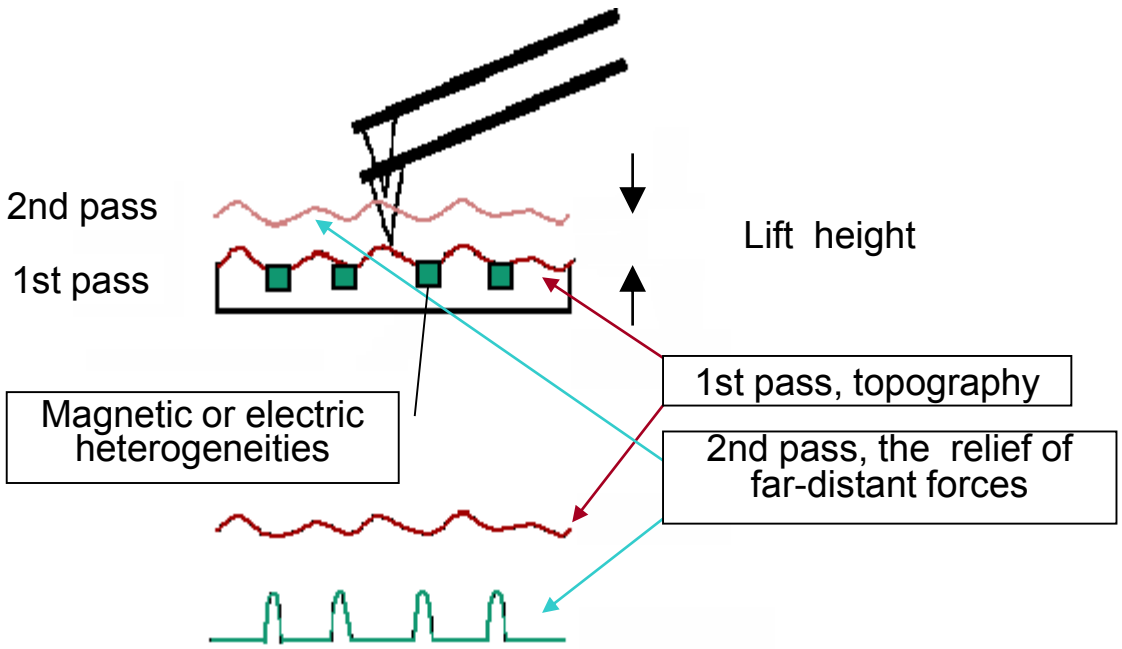
- Advantages:
  - High scan speeds
  - It is relatively easy to obtain “atomic resolution” images
- Disadvantages:
  - Due to the action of lateral forces the features in the image can be distorted
  - The forces normal to the tip-sample interaction can be high in air due to capillary forces
  - The combination of lateral forces and high normal forces can result in reduced spatial resolution and may damage soft samples (i.e. biological samples, polymers) due to scraping

Advantages and Disadvantages of main AFM modes

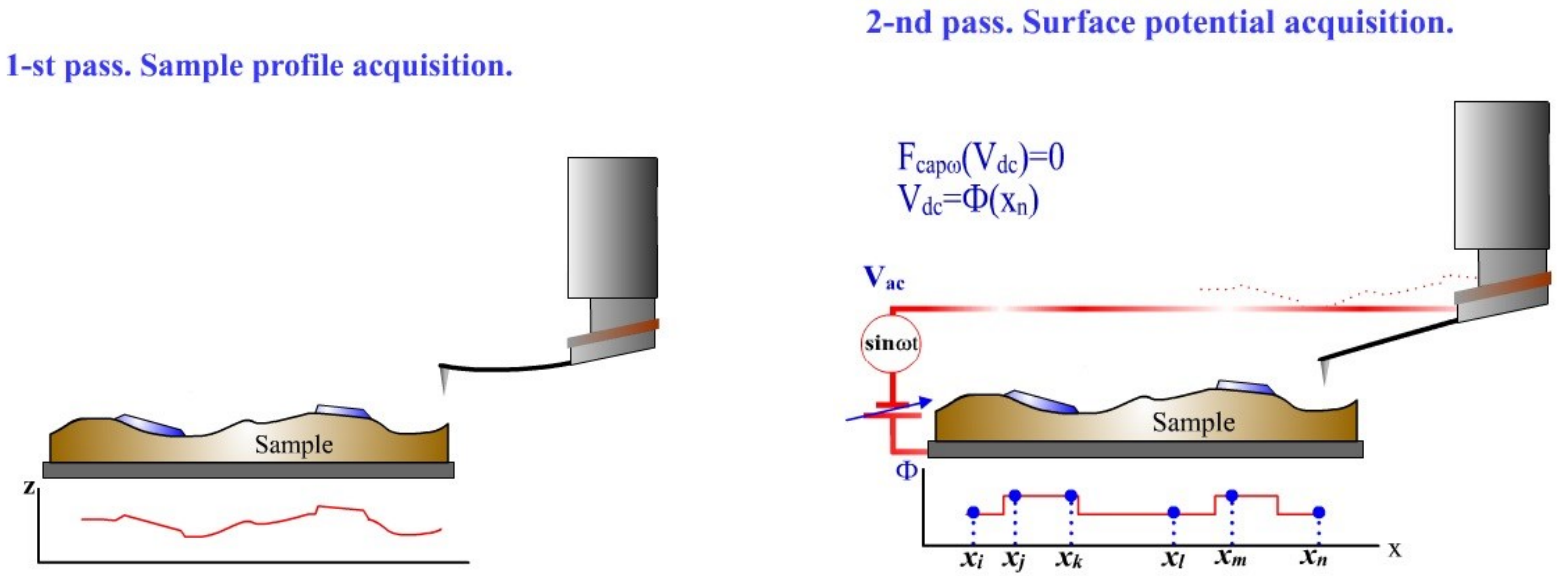
Semicontact mode

- Advantages:
  - Higher lateral resolution on most samples (1 to 5nm)
  - Lower forces and less damage to soft samples imaged in air
  - Lateral forces are virtually eliminated so there is no scraping
- Disadvantages:
  - Much lower scan speed than contact mode AFM

The principle of two-pass techniques



Two passes are made over the sample. The first measures topography while the seconds measures a material property (magnetic, electric, etc.)

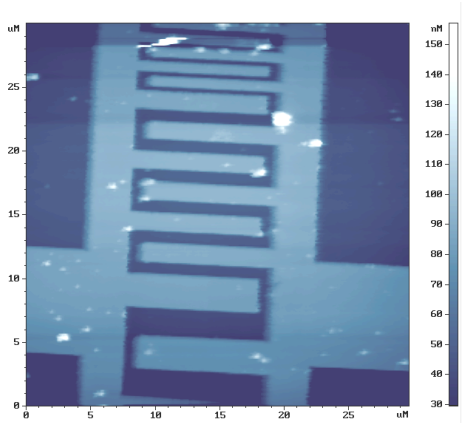


Kelvin probe force microscopy (Kelvin mode of Scanning Probe Microscopy) was invented for measuring contact potential difference between the probe and the sample. In the first pass the topography of a single line is acquired using standard semicontact mode (mechanical excitation of the cantilever). In the second pass this topography is retraced at a set lift height from the sample surface to detect the electric surface potential **Φ(x,y).** During this second pass the cantilever is no longer excited mechanically but electrically by applying the voltage V_{tip} containing dc and ac components :

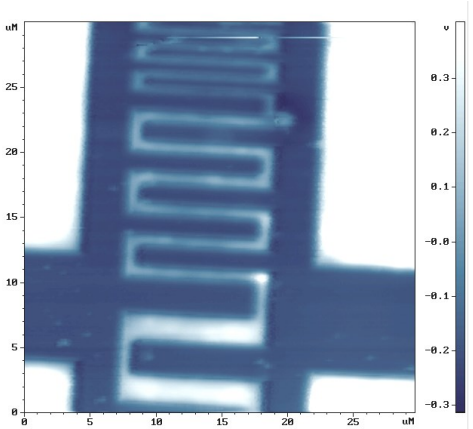
$$\mathbf{V_{tip}=V_{dc} + V_{ac}\ sin(\omega t)}$$

The second feedback then switches on and changes the dc tip potential V_{dc} until the ω component of the tip-force vanishes, this takes place when $\mathbf{V_{dc}=\Phi(x,y)}$. Obtained in such a way in each sample point value V_{dc} represents the map of surface potential.

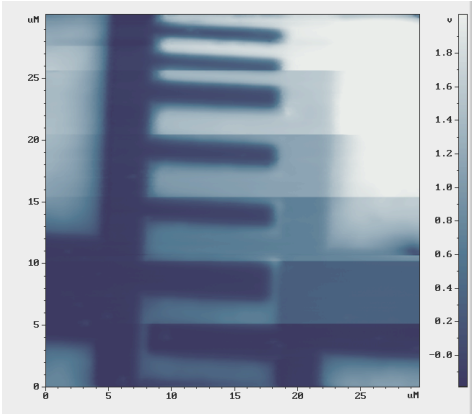
Kelvin Probe Microscopy of a finger grating



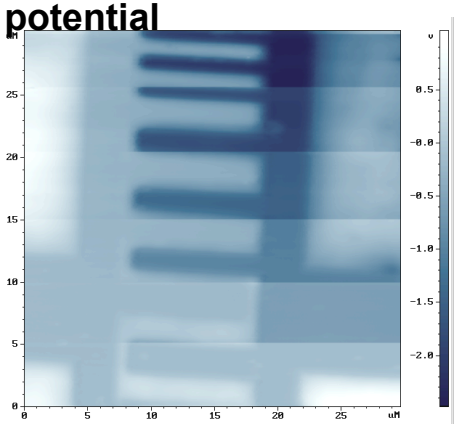
Topography



Kelvin mode, both electrodes have ground potential

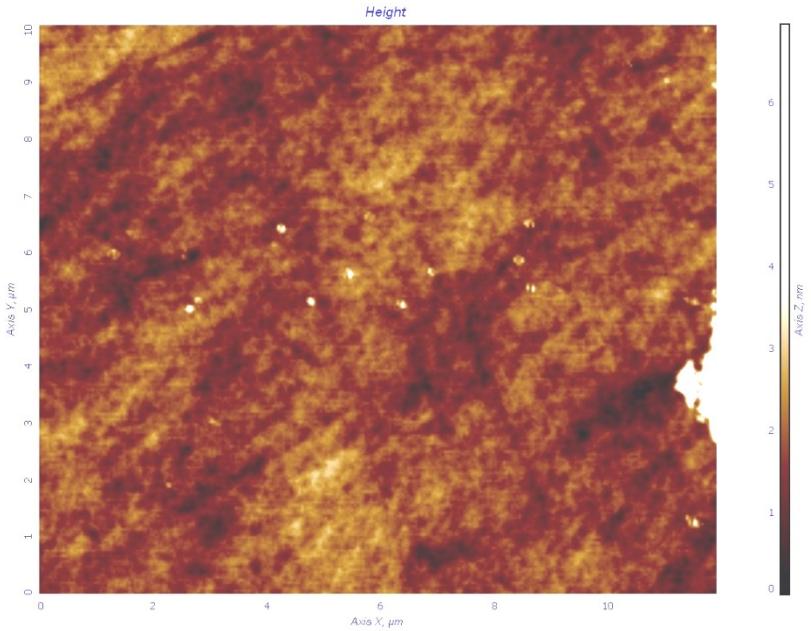


Kelvin mode, potential on the right electrode is changed with the step of +0.5V

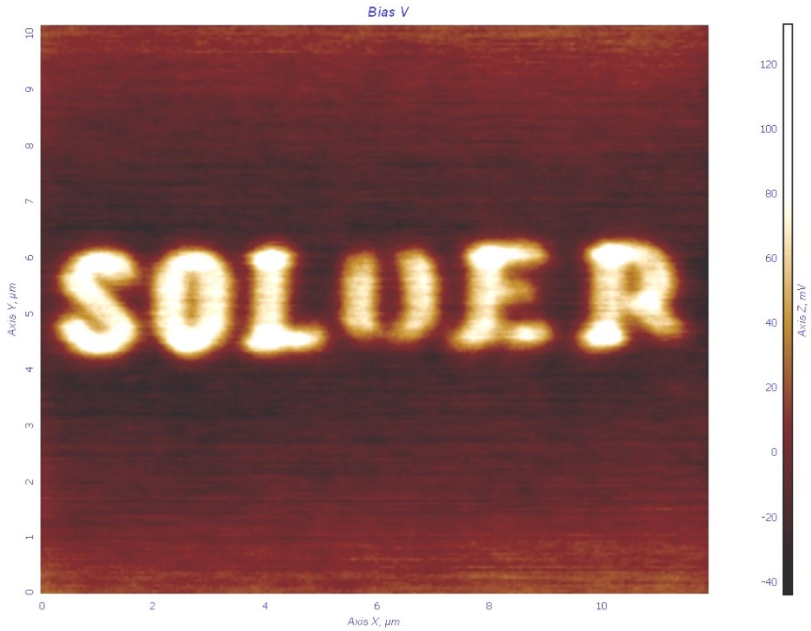


Kelvin mode, potential on the right electrode is changed with the step of -0.5V

Vector charge lithography



Topography



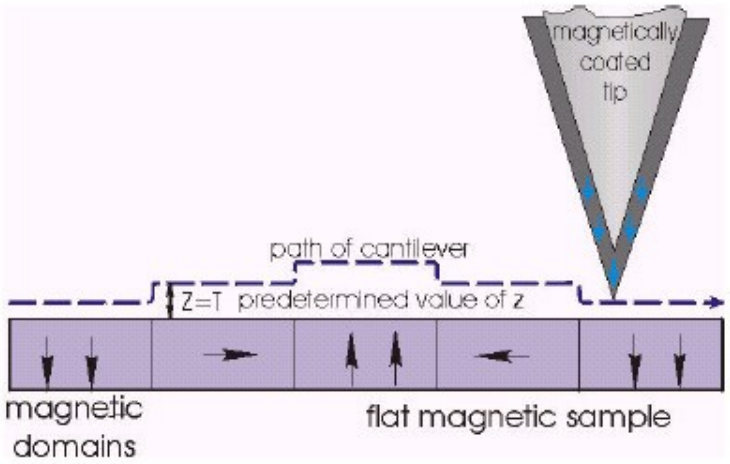
Kelvin mode

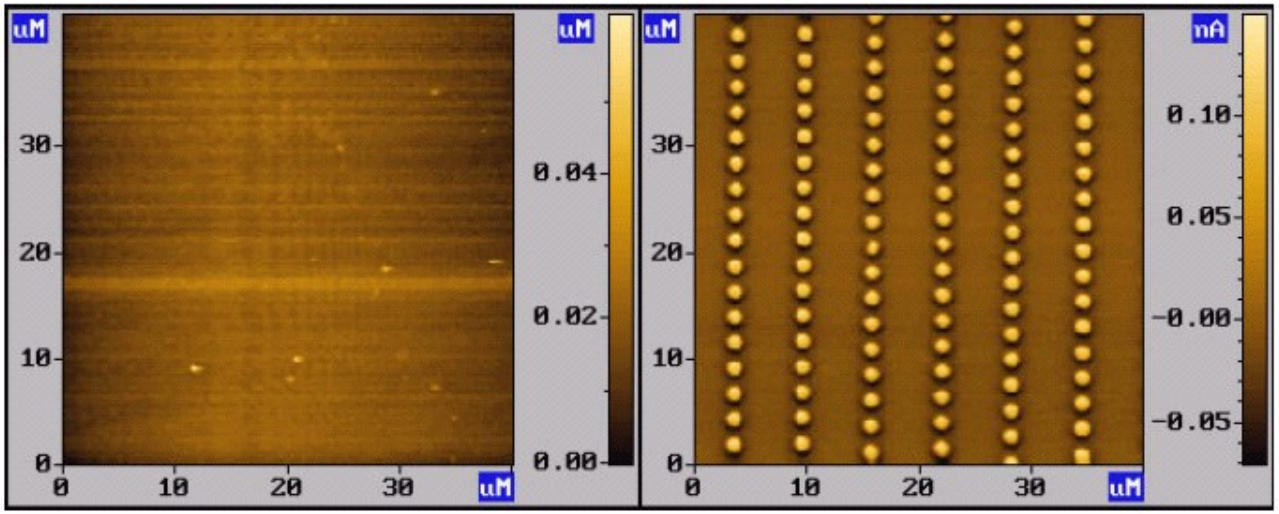
AFM images of vector charge lithography performed by a positively biased tip on GaAs surface. Apart from topography the result is clearly visible in Kelvin mode. Scanning area 10x12 µm.

Magnetic Force Microscopy (MFM)

MFM images the spatial variation of magnetic forces on sample surface. By using cantilevers coated with ferromagnetic film you can simultaneously obtain two images - topography and MFM. That is MFM measures magnetic field intensity and distribution above the sample surface using amplitude, phase or frequency shifts. MFM can image naturally occurring and deliberately written domain structure in magnetic materials.

So, MFM is useful for measuring magnetic information for storage media (disks, tape), magnets, and "soft" magnetic materials such as Permalloy and garnet films.

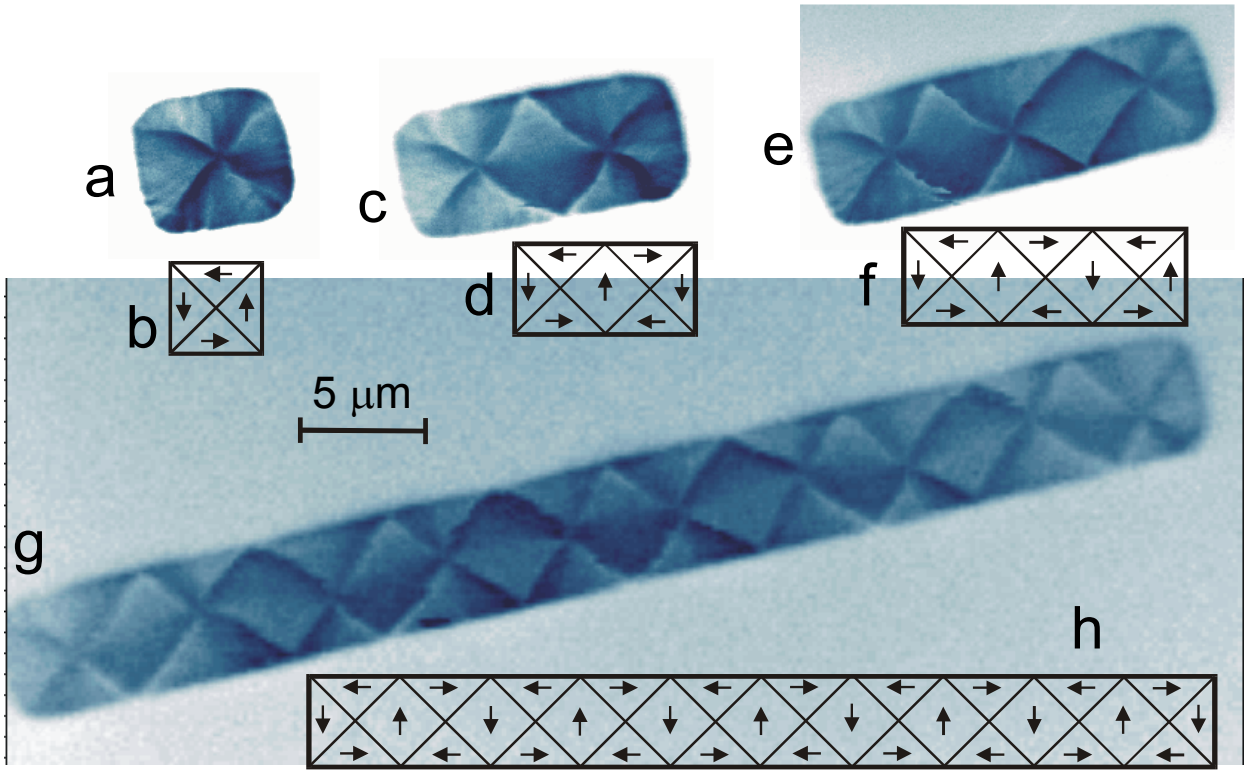




- Magneto - Optical disk.

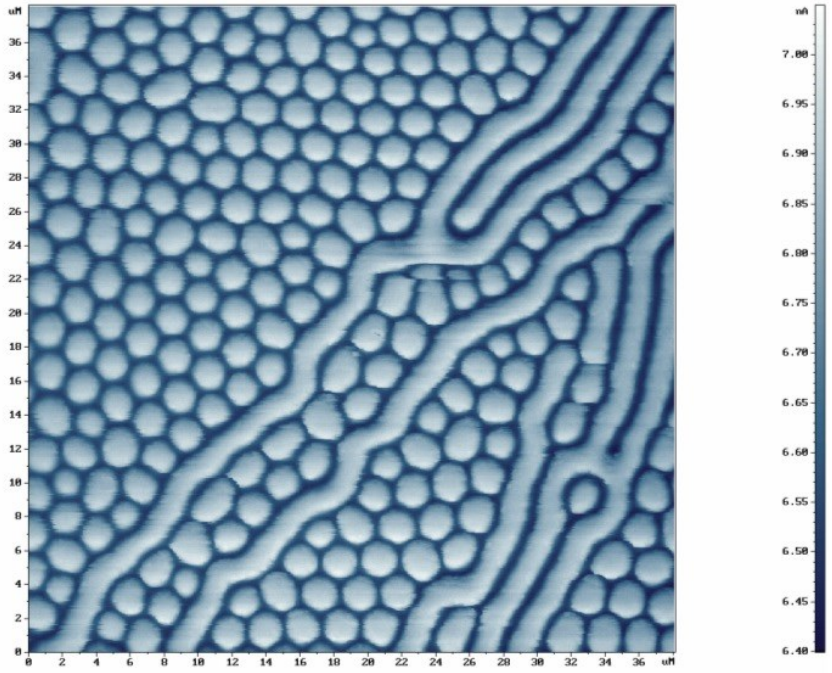
From left to right:
Topography
MFM imaging

MFM images of permalloy stripes

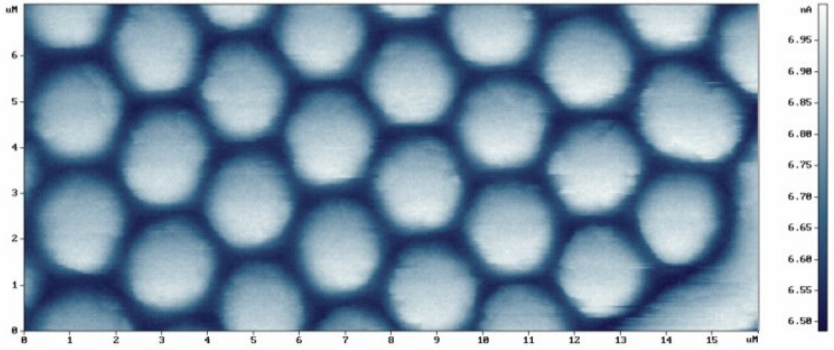


MFM images (phase) of four permalloy stripes (a, c, e, g) with a different form-factor. The interpretation of observed morphology is shown in figures (b, d, f, h) and corresponds to classical domain structure.

MFM of domain structure of a garnet film



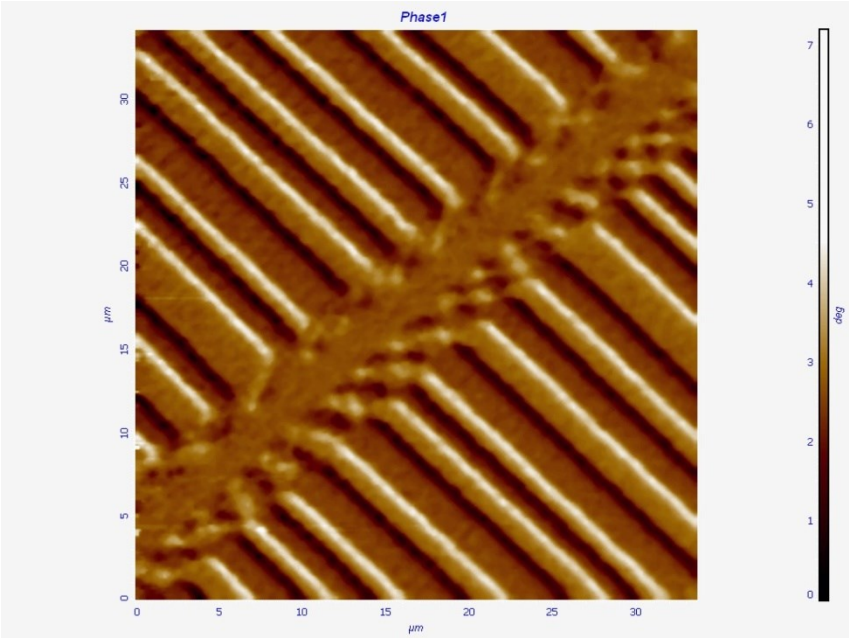
38x38 μm



7x16 μm

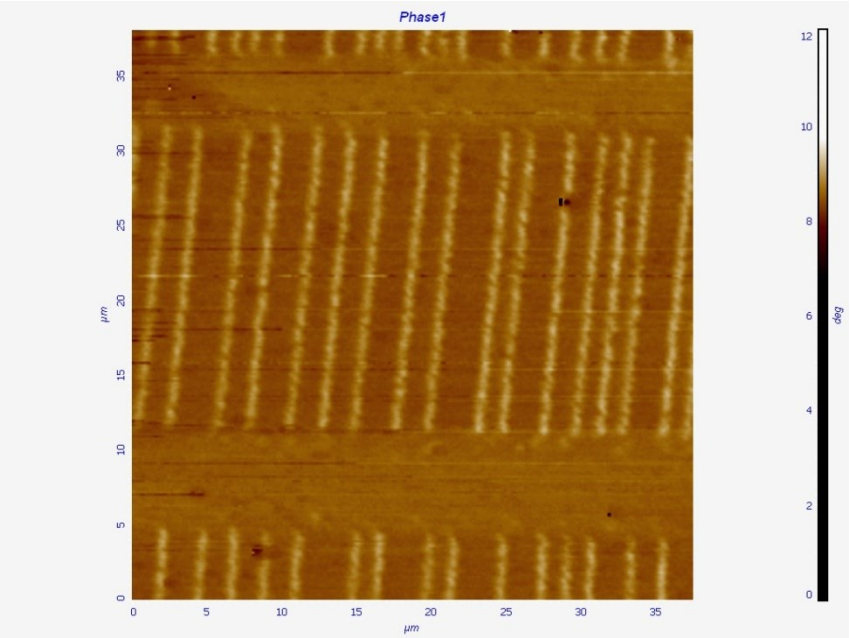
Comparison Co and FeNi coated tips

Co-coated tip



34x34
μm

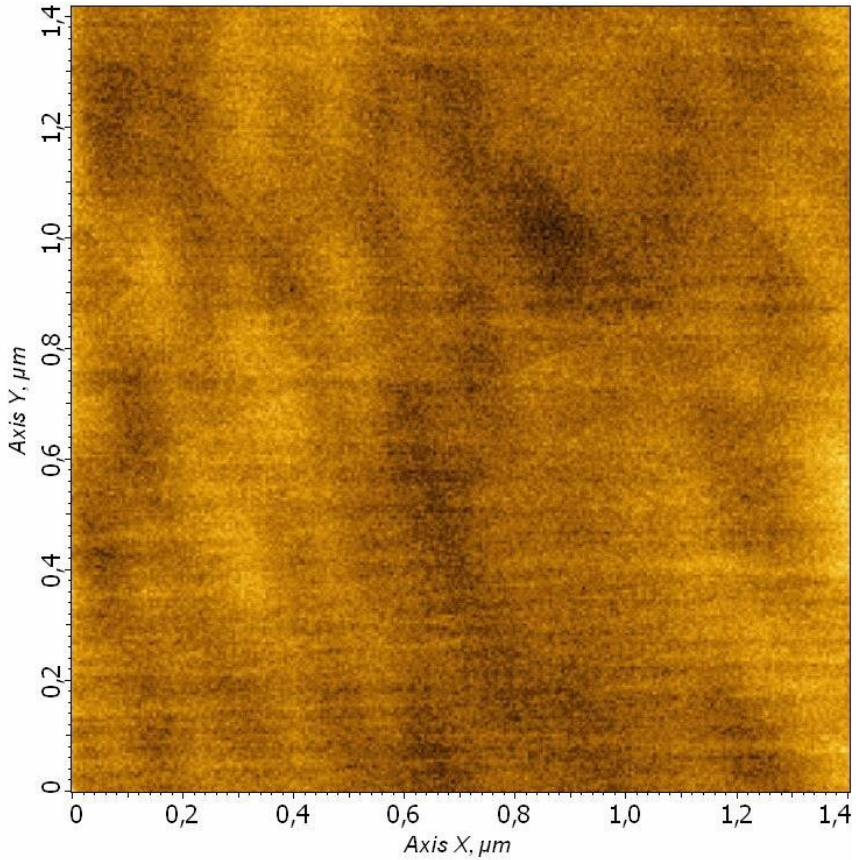
FeNi-coated tip

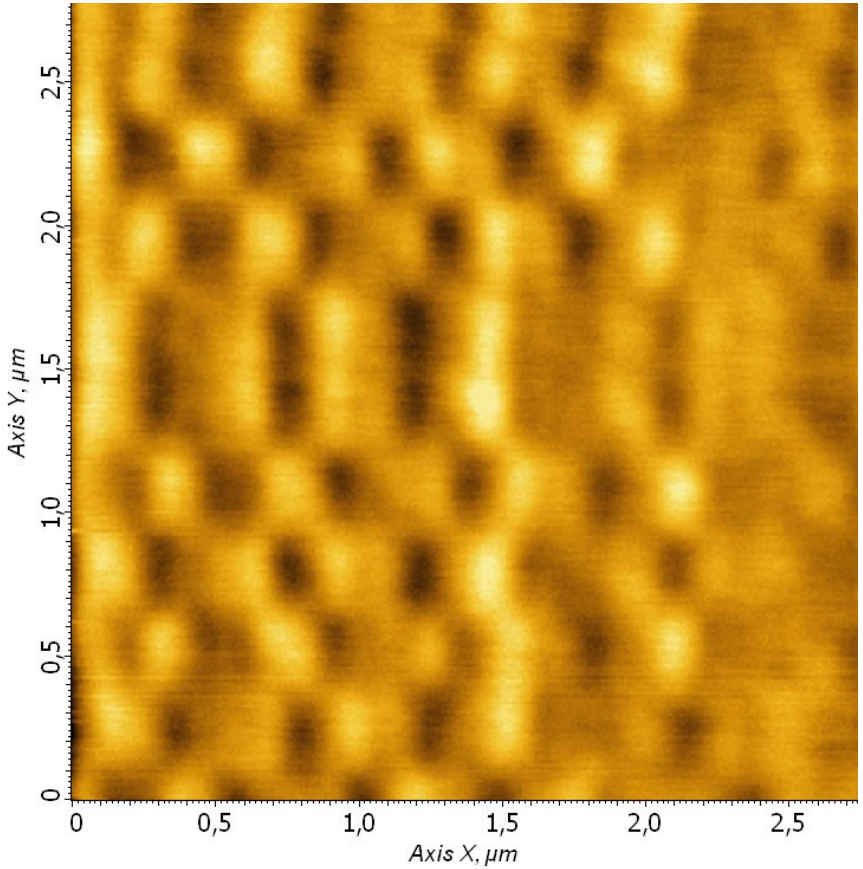


37x37
μm

MFM images of the same Hard Disk obtained by tips with different coatings: Co-coated tip (left), FeNi (right). The tip made from permalloy can show only the edges where the magnetic field of the sample domains changes the direction

MFM in vacuum – HDD of high density





In vacuum the MFM sensitivity increases proportional to the increase of the cantilever quality factor. 80Gb HDD magnetization can be clearly visualized in phase imaging mode in vacuum (5×10^{-7} Torr, right) but little is seen in air (left).

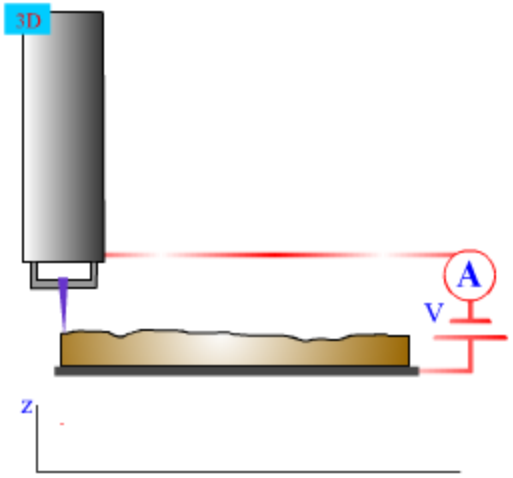
Lithography types

The first report on tip-induced oxidation of silicon was STM study by John A. Dagata in 1990.

Within the Probe NanoLaboratory NTEGRA the following kinds of lithographic surface modifications are available:

- STM Lithography
- AFM Lithography – Scratching
- AFM Lithography – Dynamic Plowing
- AFM Oxidation Lithography
- Constructive Lithography
- AFM Charge (Electrostatic) Lithography

STM Lithography



STM Lithography on LB film.
STM image of three monolayer LB film after local exposure to three electric pulses.
Scan size: 256 x 256 nm

STM can modify the surface and material can be transported from the tip to the sample and back. If these actions can be performed in a controlled way widespread possibilities would arise: information storage devices, nanometer patterning technique, manipulations of big molecules and individual atoms, building of small devices.

The most straightforward way to machine a surface by STM is by pushing the tip into the surface. This can result in a hole, but the tip can be damaged.

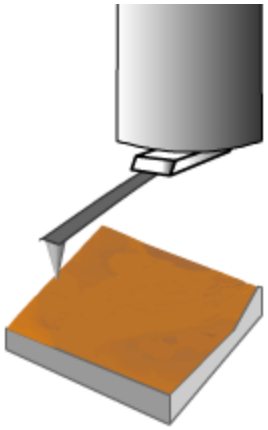
More protective is the use of a current pulse applied to STM tip. The sample surface under the tip can be locally melted and evaporated

AFM Lithography

AFM enables the direct machining the sample surface by means of AFM cantilever tips. This can be achieved in two ways:

- Static Plowing (Scratching)
- Dynamic Plowing.

AFM Lithography-Scratching

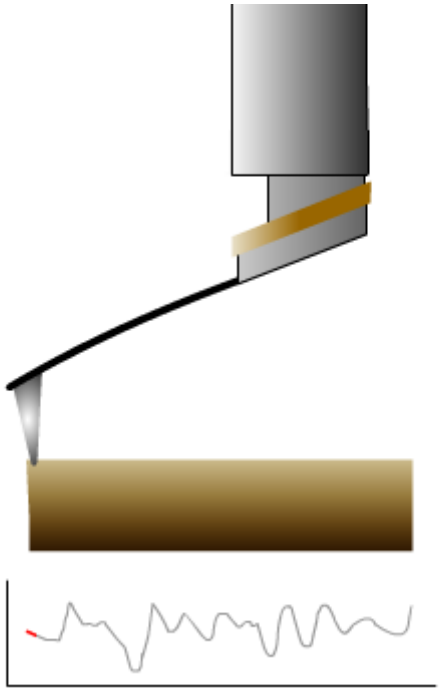


AFM Scratching Lithography
Polished Al surface scratched with 100 nN/m cantilever.
Scan size: 1.6 x 1.6 um

Plowing is a well-known technique used since the earliest days of agricultural cultivation. By scaling this tool down in size to a few nanometers and combining it with conventional scanning probe techniques, one can facilitate nanolithography with nanometer resolution.

In the more common AFM scratching techniques, the tip is scanned under strong loading forces to remove the substrate or resist. This technique utilizes the principle of plowing in the same way as the traditional tool: material is removed from the substrate in a well-defined way, leaving behind deep trenches with the characteristic shape of the plough used.

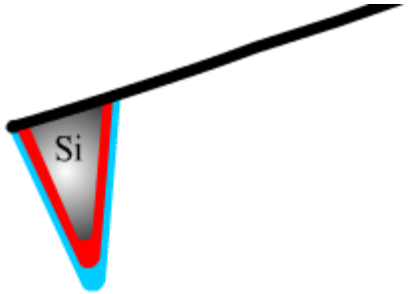
AFM Lithography-Dynamic Plowing.

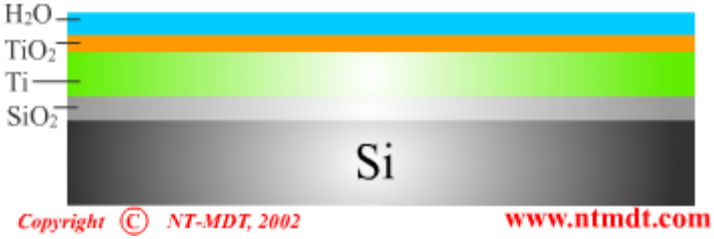


By Dynamic Plowing Lithography (DPL) the surface is modified by indenting it with a vibrating tip in the AFM semi-contact mode.

This method provides a lithography technique that is nearly free from problems due to cantilever torsion and permits to image the modified surface without any further modification.

AFM Oxidation Lithography.





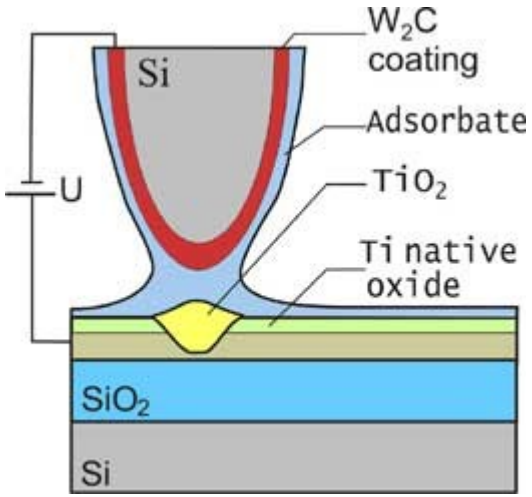
AFM Anodic Oxidation Lithography is a kind of the AFM Voltage Lithography.

With the use of the Voltage Lithography not only geometrical properties but also local electrical properties of the sample surface can be changed.

The voltage application to a conductive cantilever stimulates oxidation electrochemical processes in a gap between the probe and the surface in a thin surface layer which thus can be modified.

AFM Oxidation Lithography.

General concept of a local anodic oxidation method

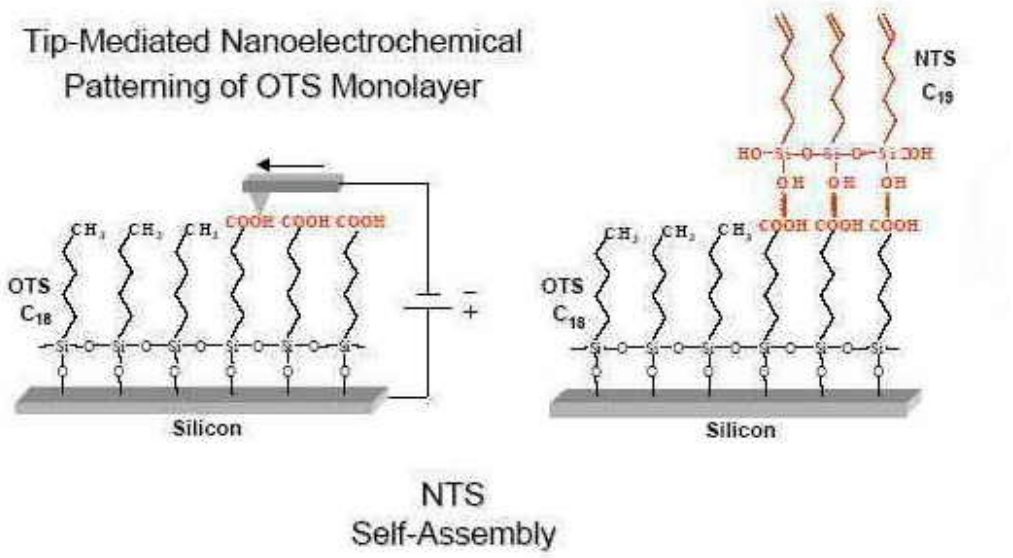


$Ti + H_2O \Rightarrow TiO_2 + 4H^+ + 4e^-$

In air or any humid atmosphere a probe and sample surface are typically covered by a thin film of absorbed water. While the tip approaches sufficiently close to the surface this layer comes in a contact with a tip and forms an electrolyte bridge by capillary action. With a negative voltage applied to the tip an electrochemical reaction takes place on the titanium surface.

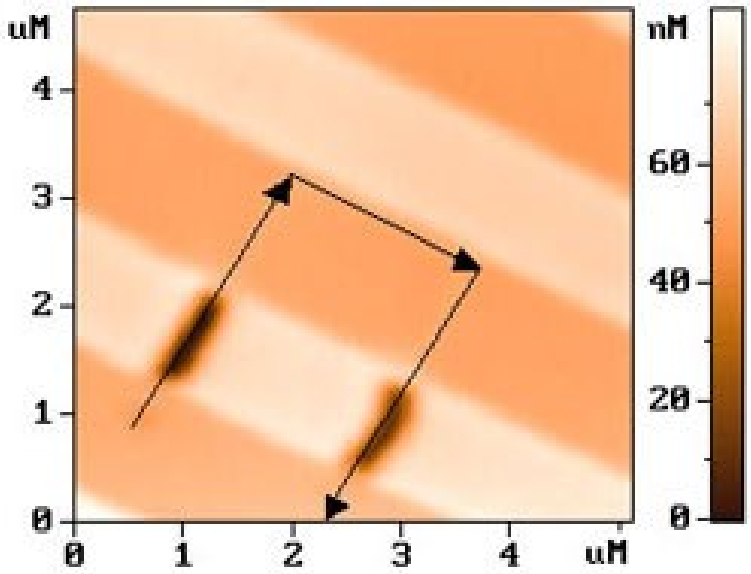
Constructive Lithography

Constructive Nanolithography is a generation on a silicon organic monolayer nano-patterns of any predefined geometrical configuration, via nano-electrochemical surface transformations carried out with the assistance of a conductive AFM tip.



Site selective self-assembly of a second monolayer (NTS) on a monolayer template pattern defined by the tip-mediated local oxidation of top -CH_3 functions of a base monolayer (OTS) to -COOH. Further chemical modification of either the tip affected OTS sites or the exposed double bond functions of NTS can be used to develop the initially inscribed chemical information, so as to generate a variety of other template surface functions on top of which guided self-assembly of diverse selected organic or inorganic nanoobjects can take place.

AFM Charge (Electrostatic) Lithography



Example of the charge writing on the surface of standard silicon grating by silicon cantilever

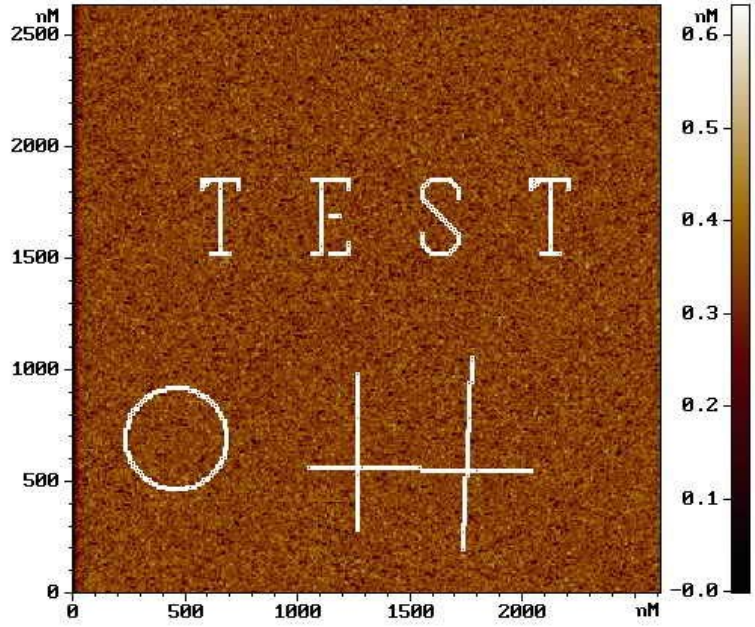
The electrization of a samples is possible with the use of cantilever.

Existence of the charge induced by a cantilever friction can explain the “apparent” holes on a silicon oxide (dark tracks). These holes appear on a topography image owing to strong electrostatic tip-sample interaction displacing the cantilever resonant peak during scanning in semi-contact mode. If electrostatic tip-sample interaction is attractive the charged areas look like holes. The repulsive interaction gives convexities on topography for charged areas.

. At using of the contact mode these charged areas are invisible. Moreover, all scan will be charged after operation in a contact mode. The charged areas can be easily visualized with Electrostatic Force Microscopy (EFM).

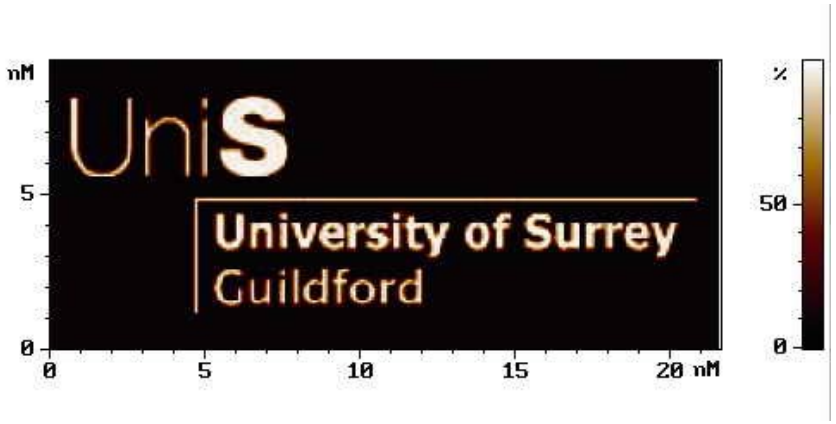
Vector and Raster lithography

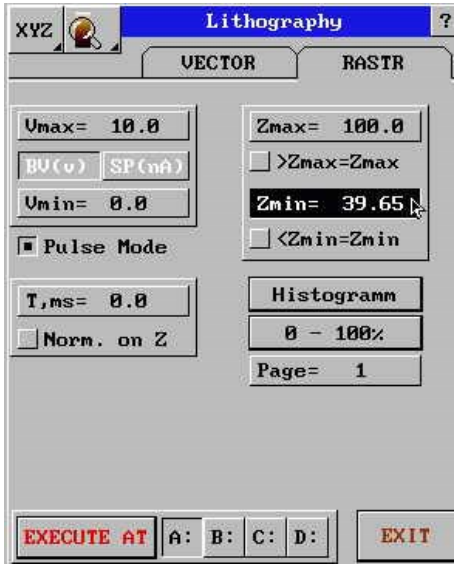
Any kinds of lithography can be performed in the vector lithography or in the raster lithography.

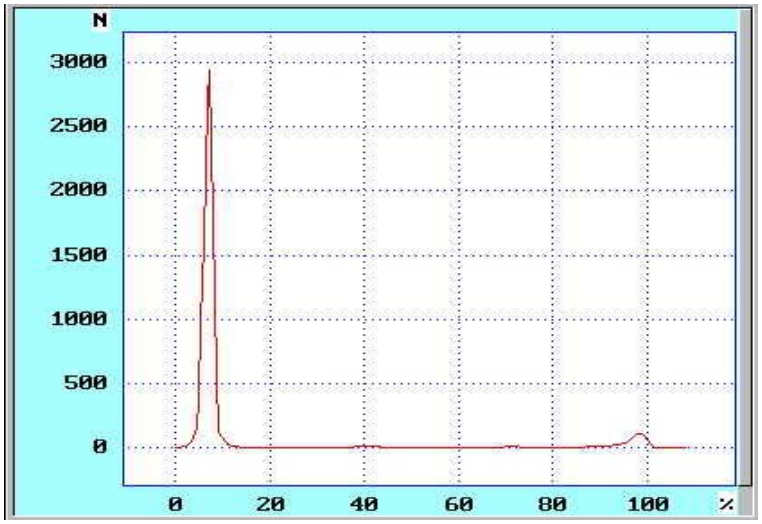


In the Vector lithography the software provides a set of commands that permit us to write lines of arbitrary length and direction with the pre-defined scan rate and the oscillation amplitude of a force or applied voltage.

Vector and Raster lithography

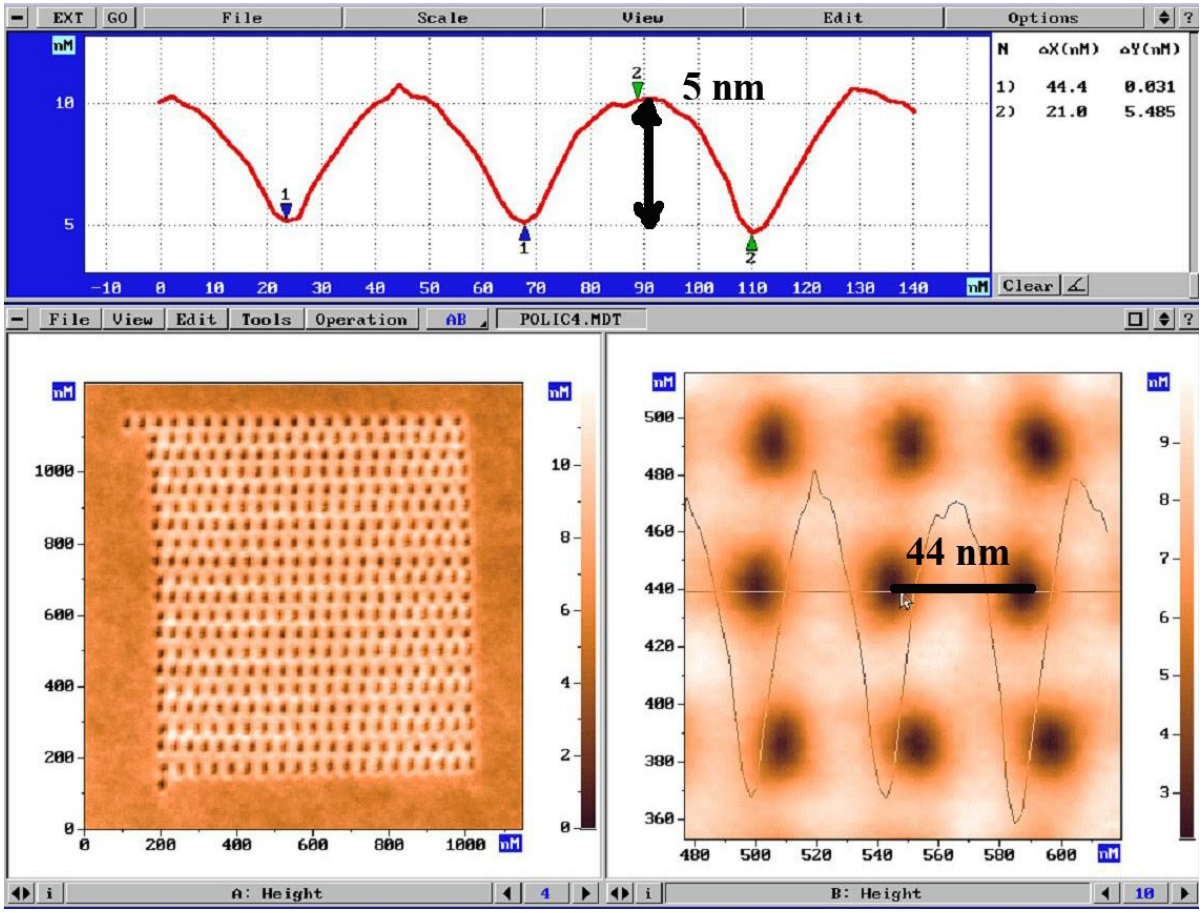






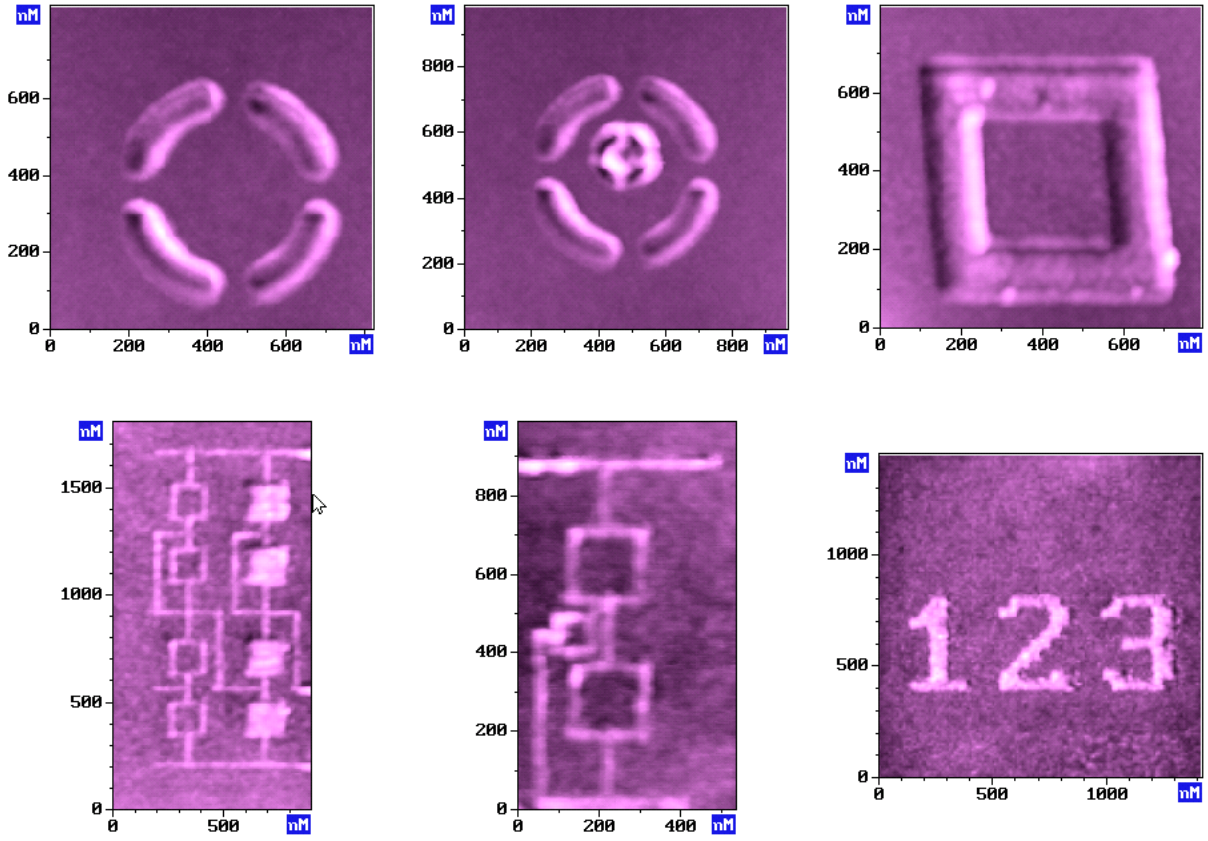
The raster lithography is synchronized with common raster scanning. The pattern can be drawn independently on a lithography itself with a use of any pixel-oriented graphical software.

Example of Force Lithography



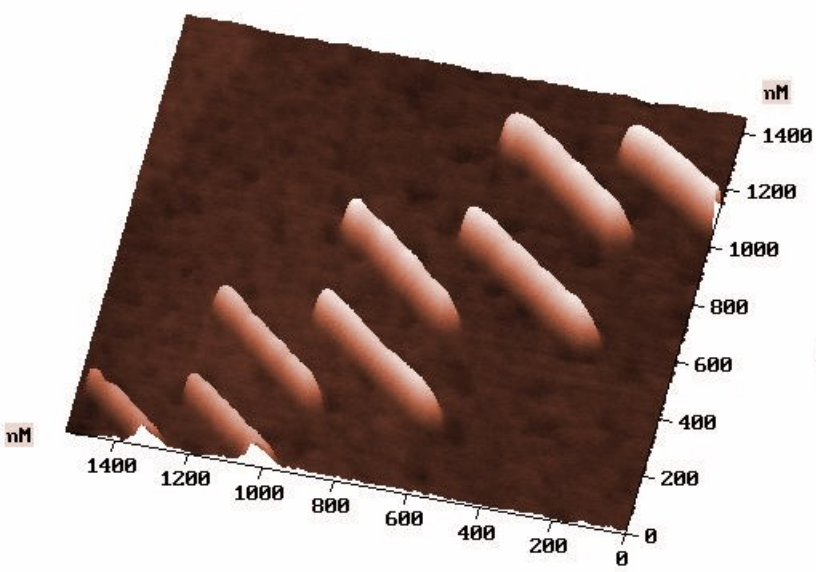
Example of bit memory matrix 20x20 on polycarbone film

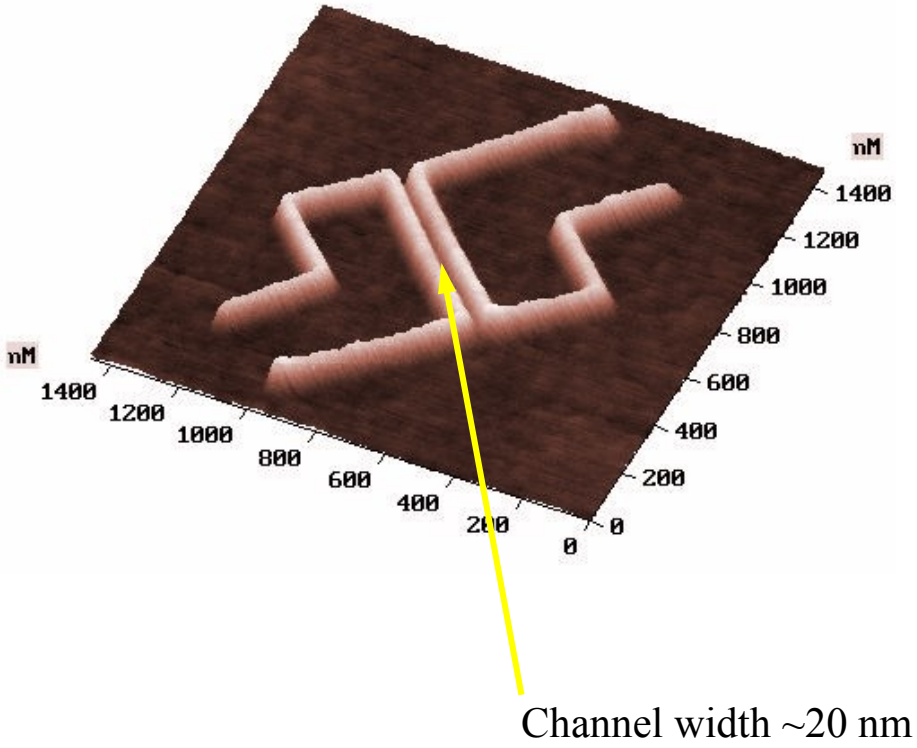
Examples of Force Lithography



Examples of an image of figures of the various form with use of a method Dynamic Plowing Lithography

Example of the vector voltage lithography

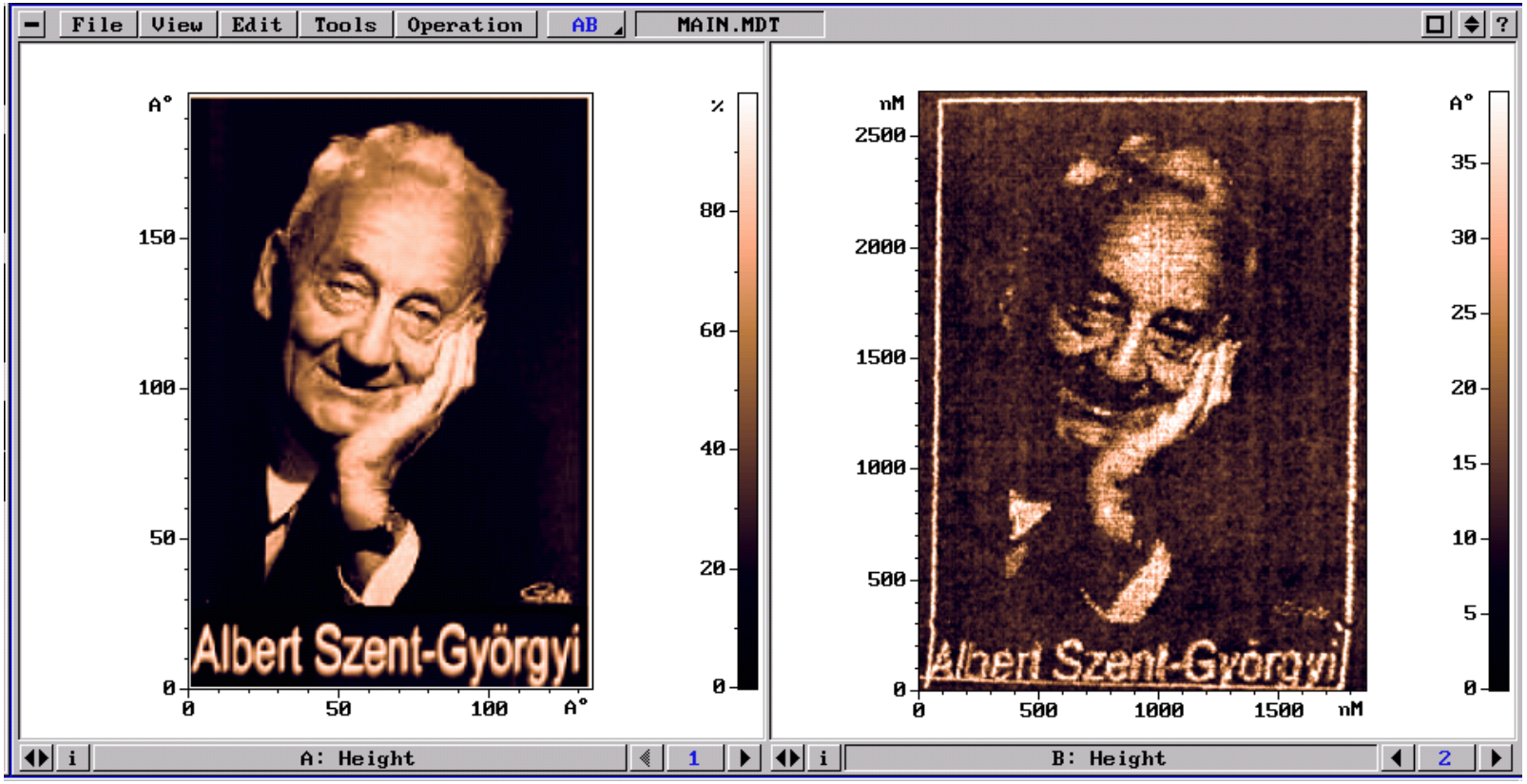




Channel width ~20 nm

Local Anodic Oxidation of Titanium film

Example of the raster voltage lithography



File with mask (left) and the AFM image (right) of titanium film after nanolithography.

Closed-Loop control

Common piezoceramics imperfections:

- onlinearities,
- hysteresis,
- creep,
- aging,
- cross coupling

Closed-loop control is used to improve the scanning performance and to expand the instrument functionality.

Scanners are equipped with low-noise capacitive XYZ sensors that provide precise positioning by measuring the exact scanner movement.

Close-loop control is very powerful method to improve scanner performance. The total nonlinearity of the system can be reduced to lesser than 1%.

Closed-Loop control

Closed-loop control as used for nanolithography, scan size 1.9x1.9 µm

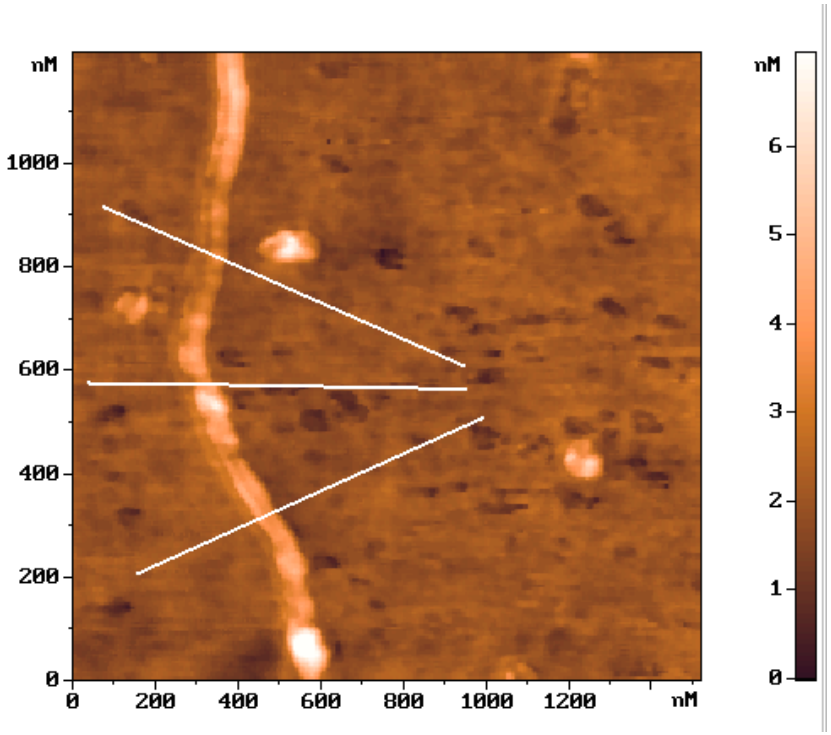


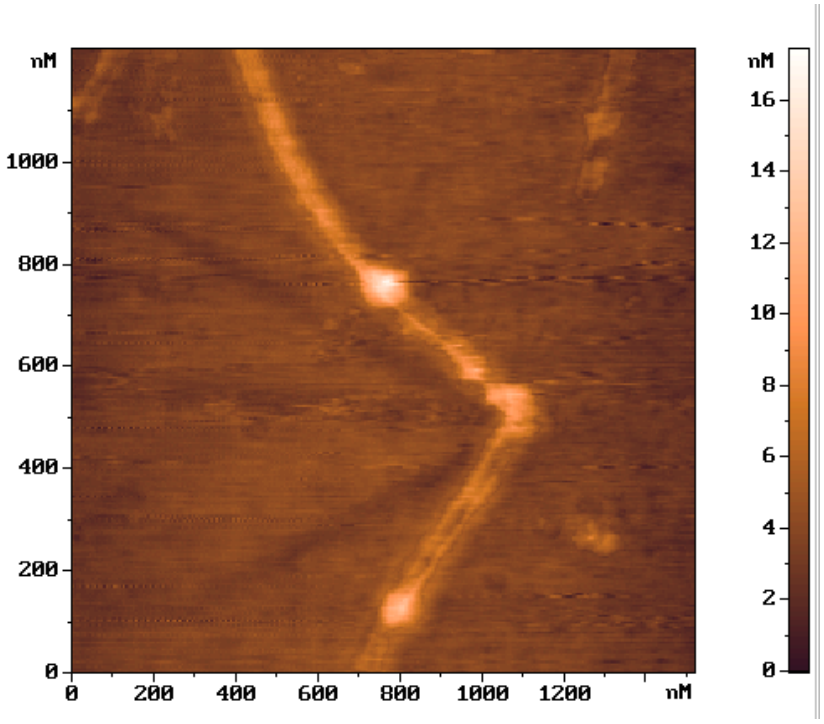
Closed-loop operation off



Closed-loop operation on

Nano Manipulation

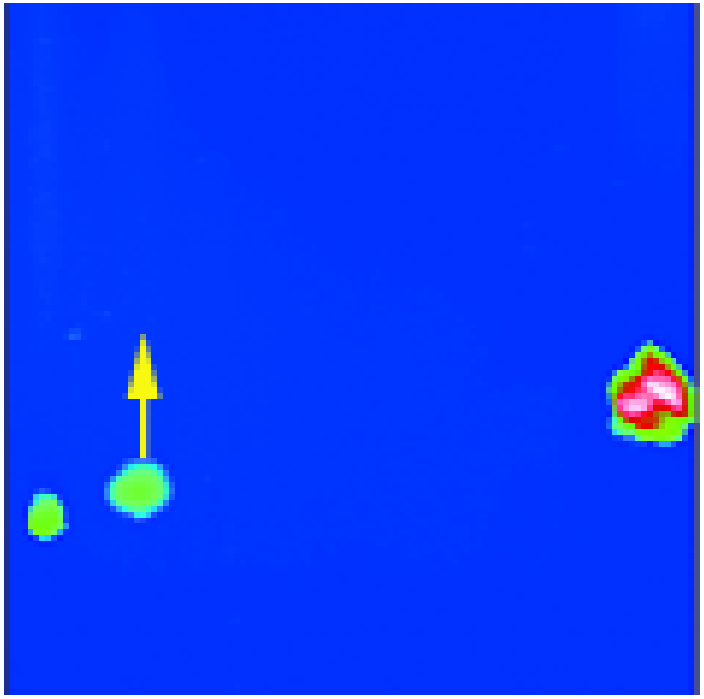




Nanotubes manipulations in the specified direction by use the Lithography option, MIEE, Institute of fullerene & nanomaterials of RAS, Russia

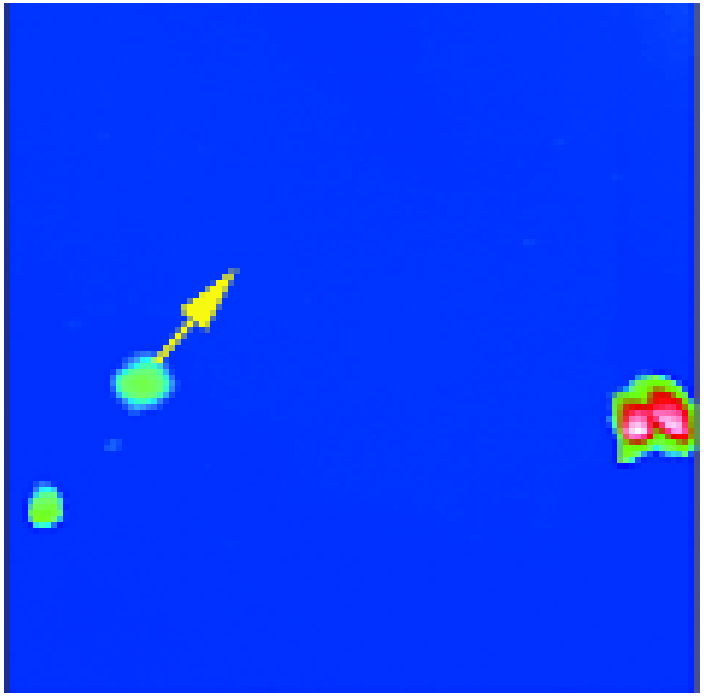
Nano Manipulation

Particle size 82nm
Reference (red) point - nanoparticle on the right.



Carbon nanoparticles (C60) on HOPG. Nanomanipulation of a nanoparticle using the Contact AFM Vector Nanolithography technique in the specified direction (see the yellow arrow).
Image size: 1,3x1,3um
The images were obtained in semicontact AFM mode.

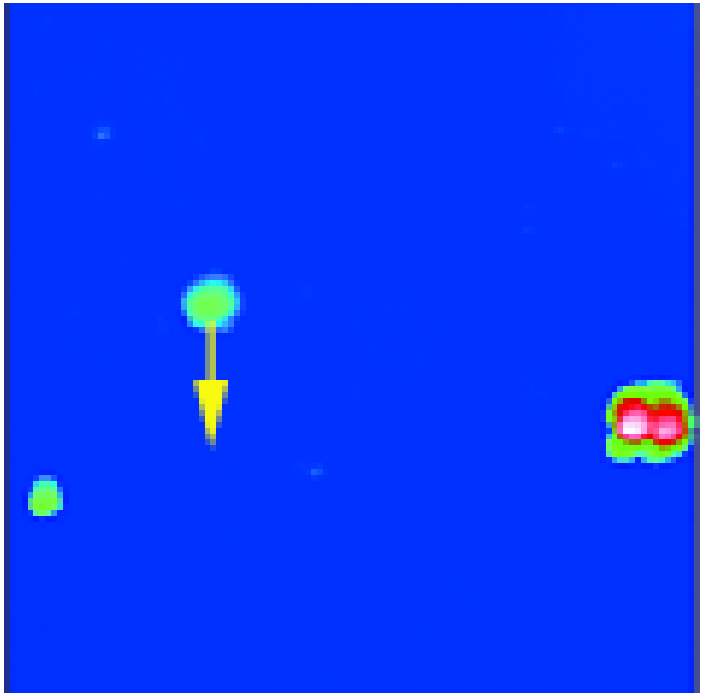
Nano Manipulation



Relocation
102nm

Carbon nanoparticles (C60) on HOPG. Nanomanipulation of a nanoparticle using the contact AFM Vector Nanolithography technique in the specified direction (see the yellow arrow).
Image size: 1,3x1,3um
Images are obtained in Semicontact AFM mode

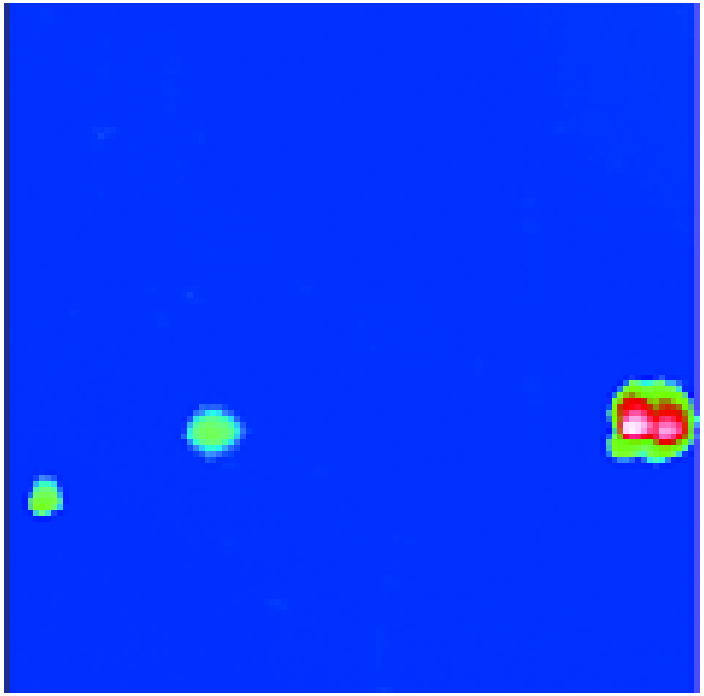
Nano Manipulation



Relocation
113nm

Carbon nanoparticles (C60) on HOPG. Nanomanipulation of a nanoparticle using the contact AFM Vector Nanolithography technique in the specified direction (see the yellow arrow).
Image size: 1,3x1,3um
Images are obtained in Semicontact AFM mode

Nano Manipulation



Relocation
138nm

Carbon nanoparticles (C60) on HOPG. Nanomanipulation of a nanoparticle using the contact AFM Vector Nanolithography technique in the specified direction (see the yellow arrow).
Image size: 1,3x1,3um
Images are obtained in Semicontact AFM mode